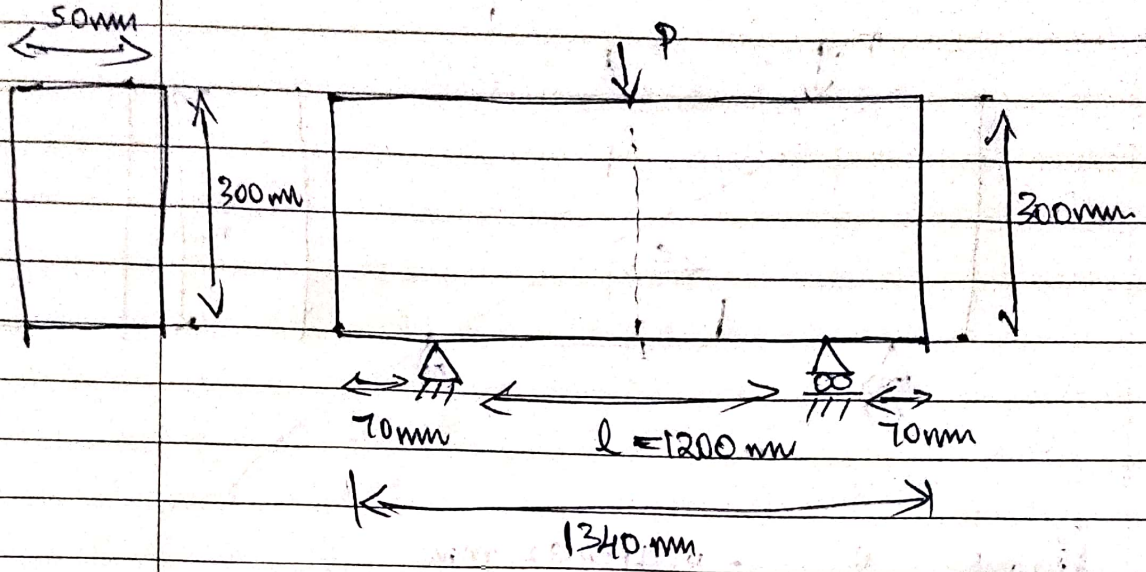
1506-26 UNNOTCHED LARGE SPECIMEN



P = 10000N (10kN) b = 50mm d = 300 mm

Total length = 1340 mm. l = 1200mm

Consider $E = 29400\ N/mm^2$ $\gamma = 0.12$

1) Support Reactions

$$R_A = R_B = P/2 \text{ (symmetry)}$$

$$\boxed{R_A = R_B = \frac{10000}{2} = 5000\ N}$$

2) Maximum Bending Moment

For a simply supported Beam with a central point load,

$$M_{max} = \frac{PL}{4}$$

$$M_{max} = \frac{10000 \times 1200}{4} = 3 \times 10^6\ N\,mm$$

Occurs at midspan

3.) Moment of Inertia:

$$I = \frac{bd^3}{12}$$

$$I = \frac{50(300)^3}{12}$$

$$I = 112.5 \times 10^6 \text{ mm}^4$$

4.) Maximum Binding stress

$$\sigma = \frac{Mc}{I}$$

where, $c = \frac{d}{2} = 150\text{mm}$

$$\sigma = \frac{Mc}{I} = \frac{3 \times 10^6 \times 150}{112.5 \times 10^6} = 4\text{MPa}$$

$$\boxed{\sigma = 4\text{MPa}}$$

σ (Bind. stress max). $= 4\text{MPa}$

5.) Midspan deflection

For centre loading, $\delta = \frac{PL^3}{48EI}$

$$\delta = \frac{10000 \times 1200^3}{48 \times 29400 \times 112.5 \times 10^6}$$

$$\boxed{\delta = 0.1088 \text{ mm}}$$

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6.) Shear force

Maximum shear:

$V_{max} = 5000N$

At the supports.

Comparison:

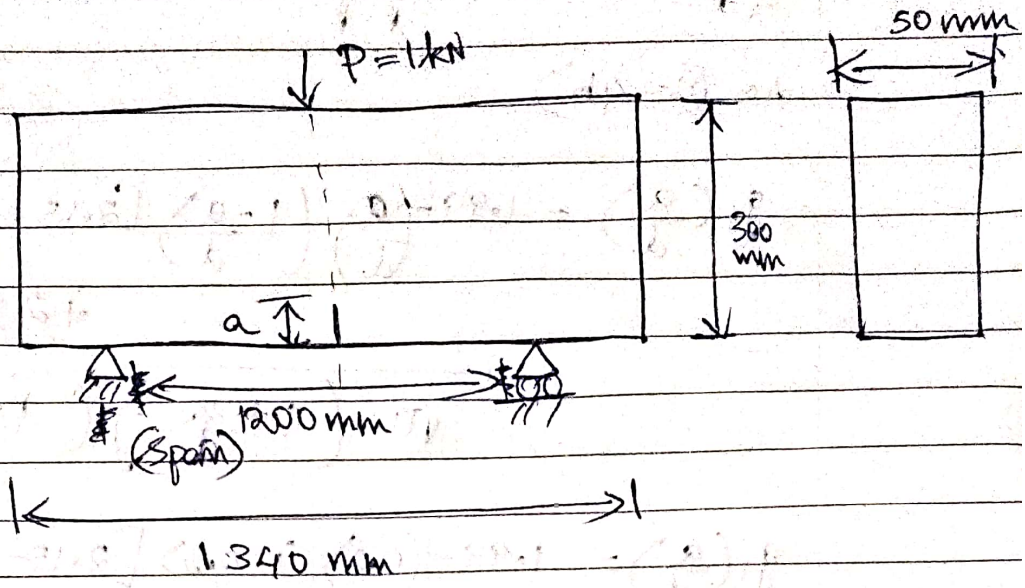
Parameter	Manual	Abaqus
Left Reaction	5000N	5000N
Right Reaction	5000N	5000N
Max Bend. stress	4MPa	3.84 MPa
Midspan deflection	0.109mm	0.145 mm

The load was directly applied ~~under~~ as a concentrated nod. force. This produces local deformation effects and stress concentration near the loading point.)

(We can apply the load through a loading roller, reference point with distributing coupling or surface contact instead of a single node concentrated force).

(Cc => dist from NA to the point where stress is being calculated)

15-06-26 NOTCHED BEAM MODE-I CALCULATION



Span(l) = 1200 mm Notch depth(a) = 60 mm
Total length = 1340 mm Load (P) = 1000 N
depth = 300 mm thickness = 50 mm

Step1: Section Modulus

$$Z = \frac{bd^2}{6} = \frac{50(300)^2}{6} = 750000\ mm^3$$

Step2: Maximum Bending Moment:
For Center Load

$$M = \frac{PL}{4} = \frac{1000 \times 1200}{4} = 300000\ Nmm$$

Step 3: Normal stress

$$\sigma = \frac{M}{Z} = \frac{300000}{750000} = 0.4\ N/mm^2 = 0.4\ MPa$$

Step4: Crack Ratio

$$\frac{a}{d} = \frac{60}{300} = 0.2$$

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Step 5: Geometry factor

As (S=.4b)

$$g_1\left(\frac{a}{d}\right) = 1.99 - \left(\frac{a}{d}\right)\left(1-\frac{a}{d}\right)\left[2.15 - 3.93\left(\frac{a}{d}\right) + 2.70\left(\frac{a}{d}\right)^2\right]$$

$$\frac{1}{\sqrt{\pi}\left(1+\frac{2a}{d}\right)\left(1-\frac{a}{d}\right)^{3/2}}$$

$$g_1\left(\frac{a}{d}\right) = 1.99 - \left\{(0.2)(1-0.2)\left[2.15 - 3.93(0.2) + 2.70(0.2)^2\right]\right\}$$

$$\frac{}{\sqrt{\pi}\left(1+(2\times 0.2)\right)\left(1-0.2\right)^{3/2}}$$

$$= \frac{1.75448}{1.775570629}$$

$$= 0.98812$$

Step 6: KI (Stress Intensity factor mode I)

$$K_I = \sigma \sqrt{\pi a}\, g_1$$

$$K_I = 0.4 \times \sqrt{\pi \times 60} \times 0.98812$$

$$K_I = 5.426 \ MPa\sqrt{mm}$$

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Step 7: Mode II stress intensity factor (K_{II})

since the geometry is symmetric, the K_{II} value is (No sliding can happen.

$$K_{II} = 0$$ as the half (right/left)

Step 8: Mixed mode Parameter

$$M = \frac{2}{\pi} \tan^{-1}\left(\frac{K_I}{K_{II}}\right)$$

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Me ⇒ Mode mixity Parameter ⇒ How much of the crack tip loading is Mode I versus Mode II

$$Me = \frac{2}{\pi} \tan^{-1}\left(\frac{5.459}{0}\right)$$

$$Me = \frac{2}{\pi} \times \frac{\pi}{2}$$

$$Me = 1$$

⇒ Me = 1 ; the failure is of Mode I type